The examination is being carried out on the **following application documents**

Description, Pages

1-30 as published

Claims, Numbers

1-15 filed with entry into the regional phase before the EPO

Drawings, Sheets

1/12-12/12 as published

1 Reference is made to the following documents; the numbering will be adhered to in the rest of the procedure.

D1 WO 2019/213437 A1 (IRIDIA INC [US]) 7 November 2019

D2 US 2013/344498 A1 (MARZIALI ANDREA [CA] ET AL) 26 December 2013

2 Amendments - Article 123(2) EPC

The dependency of **claims 3, 4** (or claim 2) and **8** (any of claims 1 to 7) does not result from the dependency in the originally filed claims. The same applies for **claims 14 and 15** (any of claims 12-13, 12-14).

The applicant is requested to provide the exact passages in the originally filed description which supports such dependency.

In the absence of evidence to the contrary, said claims do not fulfil the requirements of Article 123(2) EPC.

3 Patentability - Articles 54 and 56 EPC

3.1 Document **D1** discloses:

a system for controlling movement of at least one molecule in a first fluid having a first viscosity (paragraph [0160]: slowing down the speed of the polymer, from ca. 1MHz to ca. 100-200Hz), the system comprising: a fluid region defined by at least one fluid-retaining surface (figures 1-5), wherein the fluid region is traversable by the at least one molecule; a field-responsive fluid situated in the fluid region (paragraph [0160]: electrorheological fluid), wherein the field- responsive fluid is preferably coupled to at least a portion of the at least one fluid-retaining surface (figures 1-5); and a field generator for generating a magnetic or electric field across the fluid region (paragraph [0160]: electrorheological fluid in which becomes more viscous when a voltage is applied), wherein the field generator preferably comprises a voltage source, wherein, in response to a magnitude of the magnetic or electric field across the fluid region exceeding a threshold magnitude, a viscosity of the field-responsive fluid is greater than the first viscosity (paragraph [0160]: using a medium comprising an electrorheological fluid in which becomes more viscous when a voltage is applied).

The subject-matter of **claim 1** is therefore not new (Article 54(1) and (2) EPC).

3.1.1 Document **D1** further discloses:

a system for controlling a speed of at least one molecule (paragraph [0160]: slowing down the speed of the polymer, from ca. 1MHz to ca. 100-200Hz), the system comprising: a field-responsive fluid (paragraph [0160]: electrorheological fluid) wherein a property of the field-responsive fluid is responsive to an applied magnetic or electric field, wherein the property of the field-responsive fluid is preferably a viscosity of the field-responsive fluid (paragraph [0160]: electrorheological fluid in which becomes more viscous when a voltage is applied); means for holding the field-responsive fluid in a pathway of the at least one molecule (figures 1-5; paragraph [0160]: nanopore); and means for generating the applied magnetic or electric field across the field-responsive fluid (paragraph [0160]: electrorheological fluid in which becomes more

viscous when a voltage is applied; means for generating the voltage are thus directly and unambiguously disclosed).

The subject-matter of **claim 12** is therefore not new (Article 54(1) and (2) EPC).

3.2 Document **D2** discloses:

a system for controlling a speed of at least one molecule (figures 10A-10D; paragraph [0108]; paragraphs [0288]-[0291]: example 13), the system comprising: a field-responsive fluid (paragraph [0291]: magneto-rheological fluid) wherein a property of the field-responsive fluid is responsive to an applied magnetic or electric field, wherein the property of the field-responsive fluid is preferably a viscosity of the field-responsive fluid (paragraph [0291]: a magneto-rheological fluid so that the medium has a viscosity that inherently varies with the applied magnetic field); means for holding the field-responsive fluid in a pathway of the at least one molecule (figures 10A-10D); and means for generating the applied magnetic or electric field across the field-responsive fluid (paragraph [0291]: a magneto-rheological fluid so that the medium has a viscosity that inherently varies with the applied magnetic field).

The subject-matter of **claim 12** is therefore not new (Article 54(1) and (2) EPC).

3.3 Document **D1**, which can be considered to represent the most relevant state of the art, discloses (paragraph [0160]) a method of controlling movement of at least one molecule from which the subject-matter of **claim 9** differs in that:

detecting a speed of the at least one molecule through the fluid region; and

adjusting the magnitude of the magnetic or electric field based at least in part on the detected speed of the at least one molecule through the fluid region.

3.3.1 The technical effect can be seen in controlling the translocation speed of molecules through nanopores (originally filed paragraph [0008]). The problem to be solved by the present invention may therefore be regarded as how to modify or adapt **D1** to provide for the technical effect.

3.3.2 The solution proposed in **claim 9** of the present application cannot be considered to involve an inventive step (Articles 52(1) and 56 EPC), for the following reasons:

Although paragraph [0160] of **D1** does not explicitly disclose a speed detector coupled to the controller, a skilled person - when reading that "the speed of the polymer, from ca. 1MHz to ca. l00-200Hz, for example using a medium comprising an electrorheological fluid in which becomes more viscous when a voltage is applied, thereby slowing down the speed of the polymer through the nanopore" - would immediately recognise that speed detector coupled to the controller are implicit when measuring the above mentioned frequencies.

It is therefore considered that the skilled person would regard it as a normal design procedure to combine all the features set out in claim 9.

For these reasons, the subject-matter of **claim 9** does not involve an inventive step and does not satisfy the criterion set forth in Articles 52(1) and 56 EPC.

3.4 Document **D2** discloses:

a system for controlling movement of at least one molecule in a first fluid having a first viscosity (figures 10A-10D; paragraph [0108]; paragraphs [0288]-[0291]: example 13), the system comprising: a fluid region defined by at least one fluid-retaining surface (figures 10A-10D), wherein the fluid region is traversable by the at least one molecule (figures 10C-10D); a field-responsive fluid situated in the fluid region (paragraph [0291]: magneto-rheological fluid), wherein the field-responsive fluid is preferably coupled to at least a portion of the at least one fluid-retaining surface (figures 10C-10D); and a field generator for generating a magnetic or electric field across the fluid region (paragraph [0291]: means for generating the voltage are thus directly and unambiguously disclosed), wherein, in response to a magnitude of the magnetic or electric field across the fluid region exceeding a threshold magnitude, a viscosity of the field-responsive fluid varies (paragraph [0291]: the medium comprises a magneto-rheological fluid so that the medium has a viscosity that inherently varies with the applied magnetic field).

3.4.1 The subject-matter of **claim 1** therefore differs from this known **D2** in that:

a viscosity of the field-responsive fluid is greater than the first viscosity

3.4.2 There is no specific technical effect in the originally filed description linked to the fact that a viscosity of the field-responsive fluid is greater than the first viscosity. Thus in the absence of such an effect, the problem to be solved by the present invention may therefore be regarded as how to modify or adapt **D2** to provide for an alternative field-responsive fluid.

3.4.3 The solution proposed in **claim 1** of the present application cannot be considered to involve an inventive step (Articles 52(1) and 56 EPC), for the following reasons:

D1 (paragraph [0160]) discloses such an electrorheological fluid as providing the same advantages as in the present application. The skilled person would therefore regard it as a normal option to include this feature in **D2** in order to solve the problem posed.

For these reasons, the subject-matter of **claim 1** does not involve an inventive step and does not satisfy the criterion set forth in Articles 52(1) and 56 EPC.

3.5 The same reasoning applies, mutatis mutandis, to the subject-matter of the corresponding **claim 9**, which therefore is also considered not inventive in view of the combination of the disclosures fo documents **D2** and **D1**.

3.6 **Claims 2-8, 10-11, 13-15** do not appear to contain any additional features which, in combination with the features of any claim to which they refer, meet the requirements of the EPC with respect to novelty and/or inventive step, for the following reasons: **claims 2, 13**: **D1** and common knowledge of the skilled person (see the reasoning provided for claim 9); **claims 3-6**: **D1** implicit in paragraph [0160]; **claims 7-8**: **D1** either disclosed of implicit by the nanopore in paragraph [0160]; **claim 10**: **D1** paragraph [0160]; **claim 11**: **D1** implicit in paragraph [0160]; **claim 14**: **D1** paragraph [0160]; **claim 15**: **D1** paragraph [0160]: electrorheological fluid in which becomes more viscous when a voltage is applied.

4 Further deficiencies

4.1 The features of the claims should also be provided with **reference signs** placed in parentheses to increase the intelligibility of the claims (Rule 43(7) EPC). This applies to both the preamble and characterising portion.

4.2 **Claim 1** is not in the **two-part form** in accordance with Rule 43(1) EPC, which in the present case would be appropriate, with those features known in combination from the prior art **D1** being placed in the preamble (Rule 43(1)(a) EPC) and the remaining features being included in the characterising part (Rule 43(1)(b) EPC).

4.3 To meet the requirements of Rule 42(1)(b) EPC, at least document **D1** should be identified in the description and the relevant background art disclosed therein should be briefly discussed.

4.4 The applicant is kindly asked to **bring the description into conformity with the amended claims**. Subject-matter of the description and/or drawings not covered by the claims should be either **removed** or clearly **labelled as not being part of the present invention**.

The terms "embodiment", "aspect" and "invention" should only be used when referring to subject matter falling under the claim(s). If this is not the case, then these terms should be replaced by "**example not part of the present invention**". Replacement by the word "example" alone is not sufficient, because this could be interpreted as "example of the invention", i.e. it does not make it clear that it does not fall under the claimed invention.

Features that are comprised in the independent claim(s) should not be presented as optional, such as would be the case when using words like "may", "can", "preferably", "optionally" etc..

When an optional feature is inserted into the independent claims, the description of the embodiment(s) usually has to be adjusted by replacing "may" or "can" by "**is**" and removing the words "preferably" and "optionally".

Care should be taken during revision, especially of the introductory portion and of any statements of problem or advantage, **not to add subject-matter which extends beyond the content of the application as originally filed** (Article 123(2) EPC).

5 Further prosecution

It is not at present apparent which part of the application could serve as a basis for a new, allowable claim. Should the applicant nevertheless regard some particular matter as patentable, an independent claim should be filed taking account of Rule 43(1) EPC. **The applicant is requested to indicate how the subject-matter of the new claim differs from the state of the art and the significance thereof**.